

Hexagonal Cedar Timber Pergola — Master Construction Plan

Location: Saanich, BC

Style: Warm cedar timber — open-frame, mortise & tenon, no metal hangers

Visual Intent: Heavy 6x6 warm cedar posts on visible concrete footing pads; exposed 4x6 hip rafters with decorative tails; heavy 4x8 beam ring; diagonal knee braces; traditional mortise-and-tenon joinery.

Revision: 1.3.1 | May 2025

1. Project Overview

A 10'-0" point-to-point hexagonal cedar timber pergola with a fully open 4:12 pitched hip roof converging at a central 6x6 king post hub. Hip rafters extend past the beam ring as decorative tails. The heavy 4x8 beam ring is a visual and structural centrepiece. All lateral stability is provided by 12 diagonal knee braces (2 per post).

All joinery is traditional mortise and tenon secured with 3/4" white oak drawbore pegs — no metal post hangers or joist connectors.

Posts are mounted on visible concrete caisson pads with E66 standoff bases (no post-in-ground).

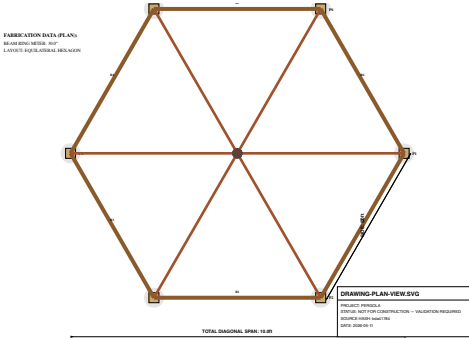
Key Specifications

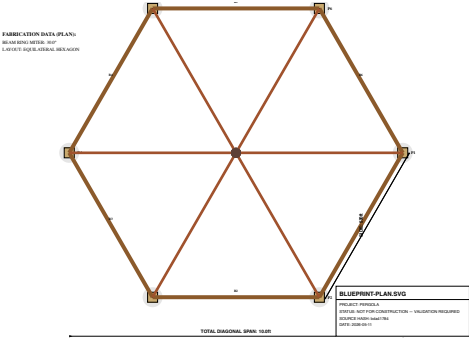
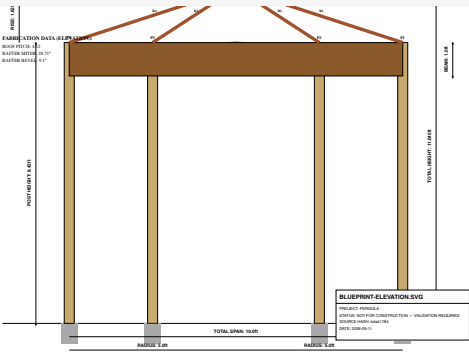
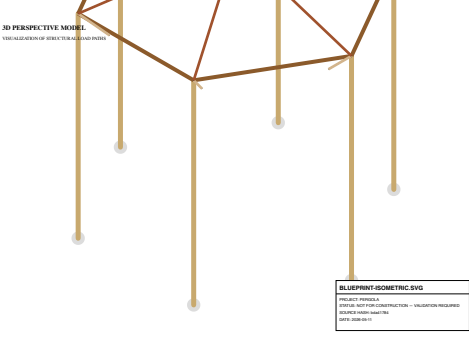
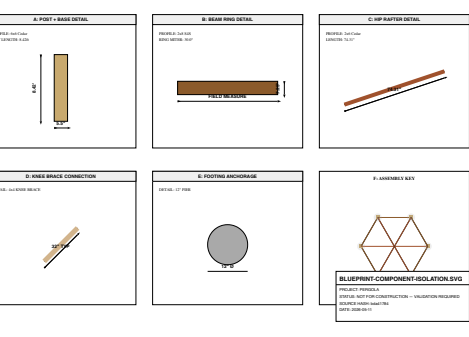
Parameter	Value
Footprint	10'-0" point-to-point diameter (5'-0" circumradius)
Side length	5'-0" (equal to circumradius for a regular hexagon)
Post spacing	5'-0" on-centre, 60° intervals
Post cut length	8'-4" (8.33 ft) — beam soffit clearance height
Total height at hub apex	10'-7" (10.601 ft)
Beam depth	7-1/4" (4x8 nominal) — heavy visible ring
Roof pitch	4:12 (18.43° pitch angle)
Hip rafter compound miter (saw swing)	28.71°
Hip rafter bevel (blade tilt)	9.10°
Beam ring flat miter	30.00° (hex geometry anchor)
Decorative rafter tail overhang	12" past beam centreline
Posts	6x6 Western Red Cedar
All members	Western Red Cedar
Finish	Penofin Transparent Cedar, 2–3 coats
Foundation	12" dia. concrete caissons, 24" deep; E66 standoff post bases
Jurisdiction / wind	Saanich BC — coastal wind ~25 psf; seismic zone 4; verify E66 hold-down with engineer

2. Architectural Drawings

All drawings produced from locked geometry (context/staging/structural-model.json).

Open SVG files in any modern browser for full zoom and inspection.

Drawing	File	Status
Plan view (top down)		✅ VALIDATED

SB-01 Plan	 FABRICATION DATA PLAN: BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 BLUEPRINT PLAN.DWG PROJECT: PERSEUS DESIGN: SHIP-POST-CONSTRUCTION - INSULATION REQUIRED DISTANCE: 10'0" (10'0") DATE: 2023-01-15 TOTAL DIAGONAL SPAN: 10'0"	Plan with post layout, dimensions, post-base callout
SB-02 Elevation	 FABRICATION DATA PLAN: BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 BLUEPRINT ELEVATION.DWG PROJECT: PERSEUS DESIGN: SHIP-POST-CONSTRUCTION - INSULATION REQUIRED DISTANCE: 10'0" (10'0") DATE: 2023-01-15 TOTAL HEIGHT: 10'0"	Elevation with compound-cut detail balloon
SB-03 Isometric	 3D PERSPECTIVE MODEL VIEW: 10'0" (10'0") BLUEPRINT ISOMETRIC.DWG PROJECT: PERSEUS DESIGN: SHIP-POST-CONSTRUCTION - INSULATION REQUIRED DISTANCE: 10'0" (10'0") DATE: 2023-01-15	Assembly isometric with sequence numbers 1–7
SB-04 Details	 A. POST - BASE DETAIL BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 B. BEAM END DETAIL BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 C. SHIP-POST DETAIL BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 D. KNEE BRACE CONNECTION BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 E. POSTING ANCHORAGE BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 F. ASSEMBLY KEY BEARING WALLS: 8'0" L-POSTS: 4x4x8, 8x8x8, 8x8x8 BLUEPRINT COMPONENT-ISOLATION.DWG PROJECT: PERSEUS DESIGN: SHIP-POST-CONSTRUCTION - INSULATION REQUIRED DISTANCE: 10'0" (10'0") DATE: 2023-01-15	4 joinery details: seat cut, tenon, post-beam, caisson

4. Structural Members & Cut List

Full machine-readable list: [../outputs/ship-blueprint/SB01-cut-list.json](#)

ID	Member	Profile	Cut Length	Qty	Notes
M01	Main posts	6x6 WRC	8'-4"	6	Square ends; seal end grain
M02	Ring beam	4x8 WRC	5'-3"	6	Opposing 28.71° miters
M03	Hip rafter	4x6 WRC	5'-4" + 12" decorative tail	6	Compound: 28.71° / 9.10°; bird's mouth at heavy beam
M04	King post hub	6x6 WRC	2'-0"	1	6× mortises at 60° intervals
M05	Knee brace	4x4 WRC	3'-0"	12	45° compound bevel both ends
M06	Purlin ring	4x4 WRC	3'-3"	6	Half-lap to rafters mid-span

Total lumber: ~490 BF Western Red Cedar (including 10% waste)

5. Joinery Details

All joints: traditional mortise and tenon, 3/4" white oak drawbore pegs. No metal hangers or joist connectors.

Joint	Detail Sheet	Mortise Size	Peg Count
Post top → beam	SB-04 Detail 3	2.125"x4.125"x3" blind	2 pegs
Rafter → hub	SB-04 Detail 2	1.625"x3.125"x3" blind	2 pegs
Knee brace → post and beam	SB-04 Detail 3	1.5"x3"x2.5" blind	2 pegs per end
Purlin half-lap → rafter	SB-04 Detail 1 note	Half-lap depth 1.5"	1 peg per joint

See [SB-04 component isolation sheet](#) for full dimensioned details.

6. Compound Cut Reference

Test on scrap before every production run. These angles were computed by `geometry_engine.py`.

Cut	Miter (fence)	Bevel (blade)	End
Hip rafter seat cut	28.71°	9.10°	Top/hub end
Hip rafter tail	0°	9.10°	Tail/overhang end
Ring beam ends	28.71°	0°	Both (opposing)
Knee brace ends	45°	0°	Both ends

7. Foundation

Element	Spec
Caisson diameter	12"
Caisson depth	24" below grade
Frost depth (Saanich BC)	18" nominal
Anchor	5/8"x10" J-bolt galvanised, set in wet concrete
Post base	Simpson E66 standoff (no post-in-ground)
Concrete	Quikrete 80 lb High-Strength, 2 bags per caisson

Permit note: verify setback from property lines with City of Saanich (250-475-5671) before digging.

Seismic zone 4 — confirm E66 hold-down spec with an engineer.

8. Bill of Materials Summary

Document	File
Detailed cut list (JSON)	../outputs/shop-blueprint/SB01-cut-list.json
Lumber purchase list	lumber-purchase-list.md
Budget estimate (CAD)	budget-estimate.md

Budget range: \$3,090 – \$4,850 CAD (DIY labour; Saanich BC April 2025 pricing)

Mid-case estimate: \$3,855 CAD

9. Assembly Sequence

Full step-by-step guide: [assembly-guide.md](#)

Phase	Task	Crew
1	Site layout + caissons	1-2
2	Post preparation + shop cuts	1
3	Erect posts + beam ring	2 min.
4	Knee braces + rafters + hub + purlins	2-3
5	Final finish + inspection	1

Estimated build time: **4-6 days** for competent DIY crew of 2 (excludes curing time).

10. Geometry Source

All angles, heights, and lengths derive from `plugins/garden-structure-designer/scripts/geometry_engine.py` .

Do not re-derive by hand — the engine handles the compound-angle trig correctly.

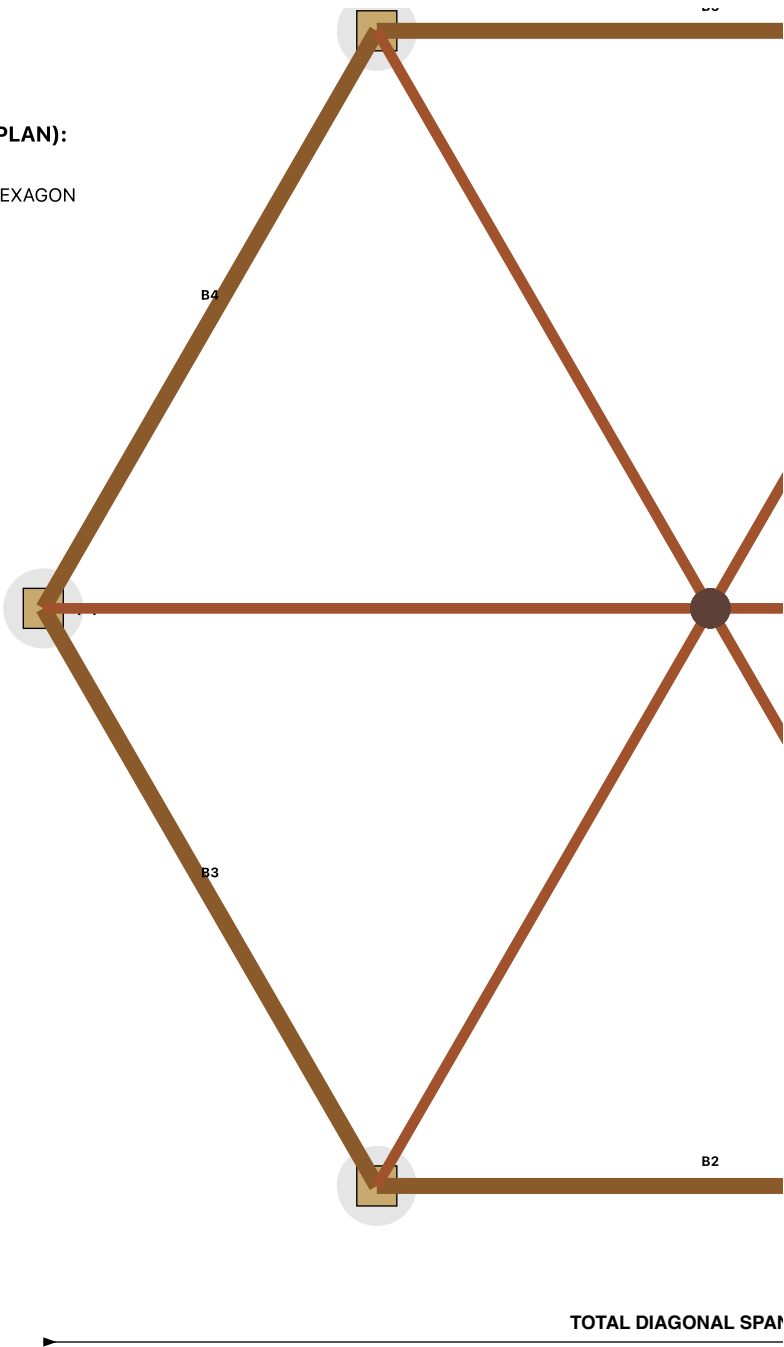
Locked model: `context/staging/structural-model.json ( _locked: true )`

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  "miter_deg": 28.71,
  "bevel_deg": 9.10,
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  "total_height_ft": 10.601,
  "rafter_overhang_ft": 1.0
}
```

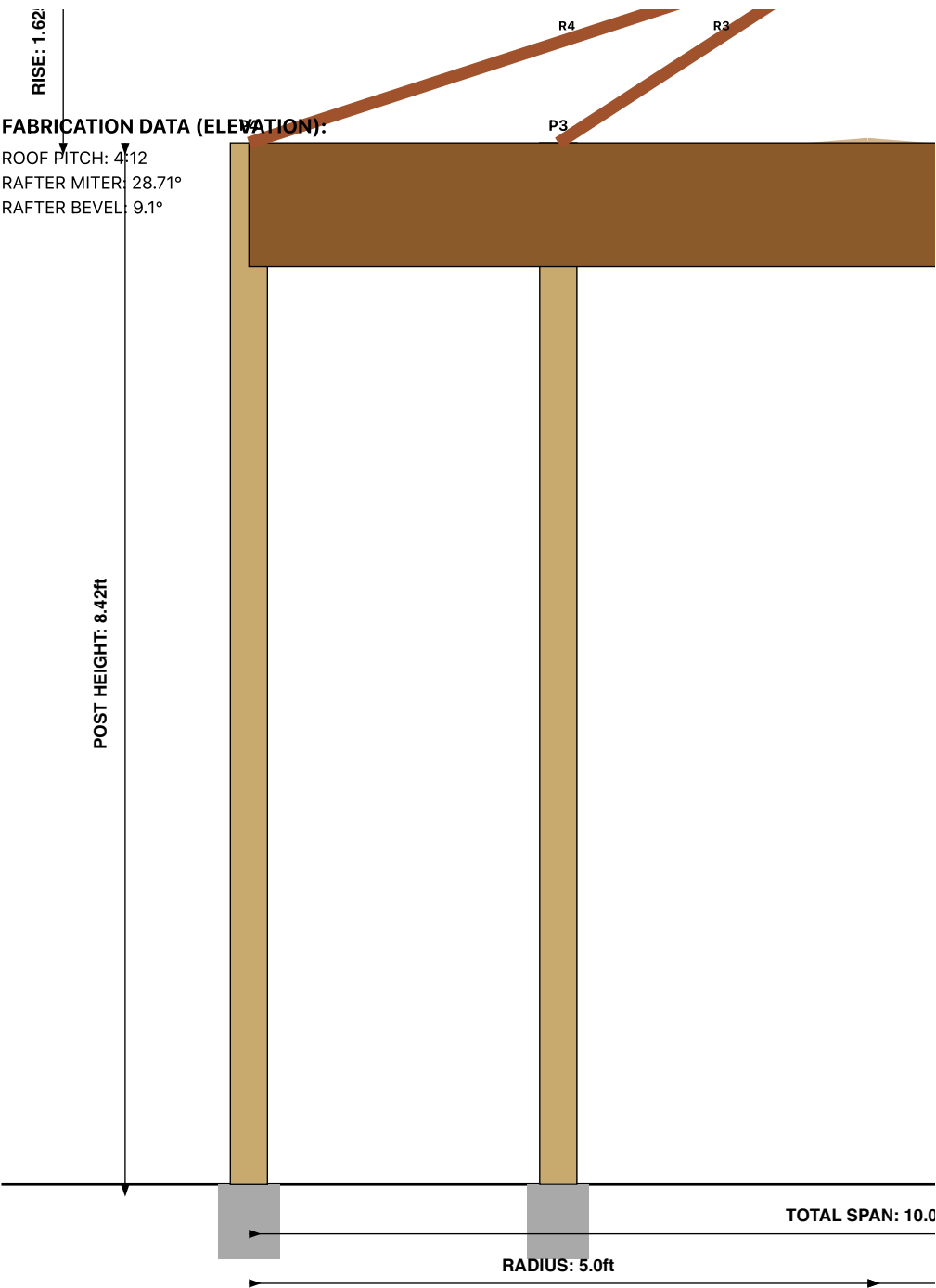
11. Full-Page Rendered Drawings

Architectural Drawings
Plan View

FABRICATION DATA (PLAN):
BEAM RING MITER: 30.0°
LAYOUT: EQUILATERAL HEXAGON

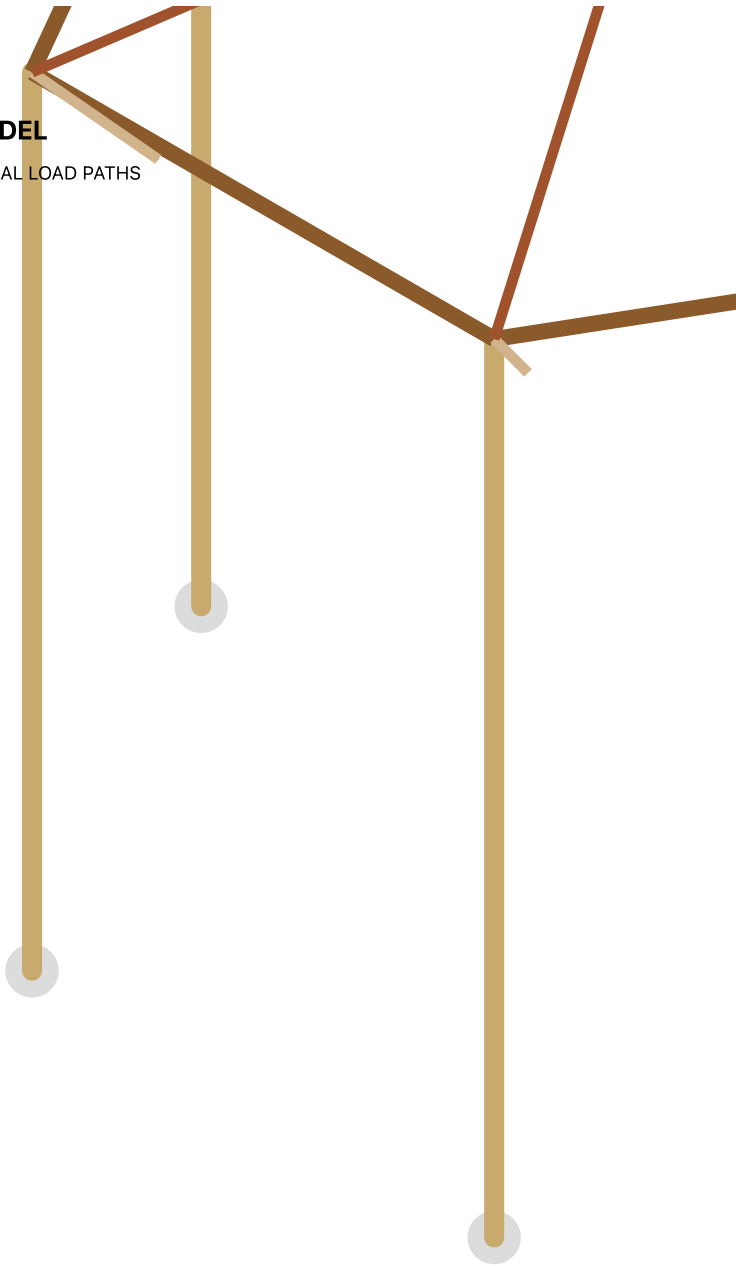


Front Elevation



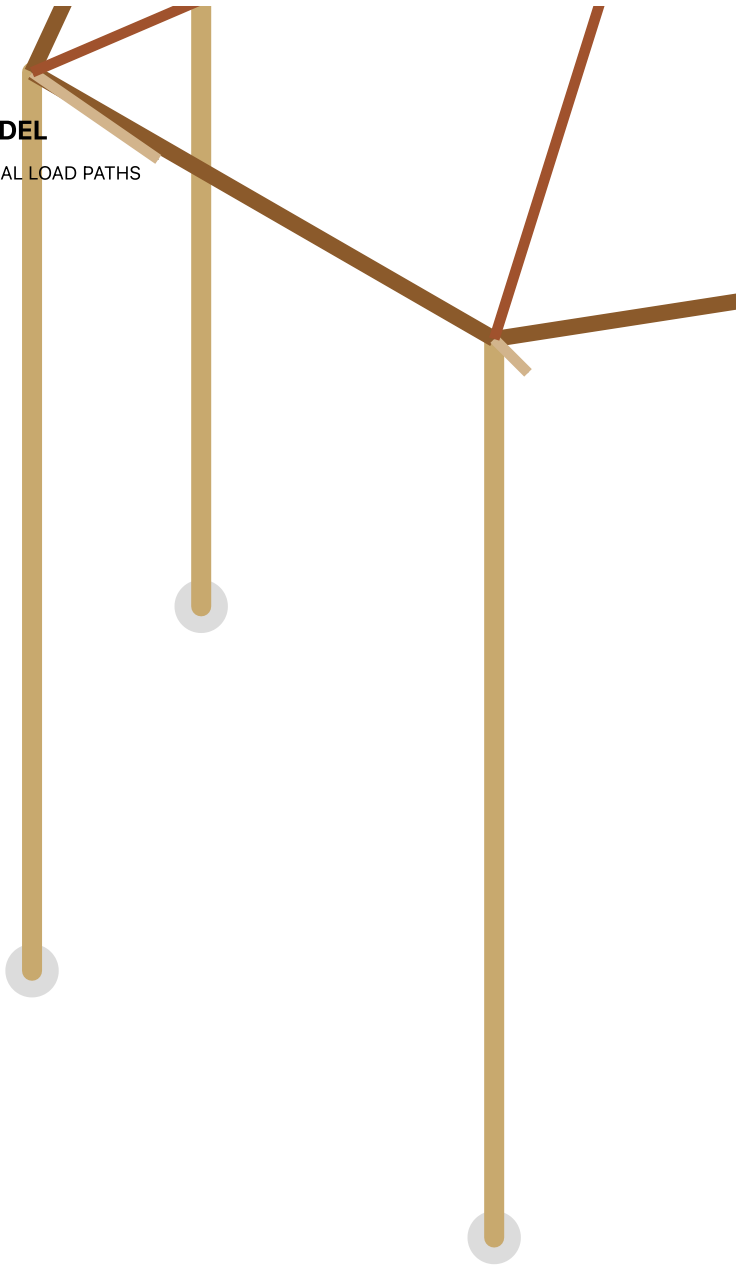
Isometric Projection

3D PERSPECTIVE MODEL
VISUALIZATION OF STRUCTURAL LOAD PATHS

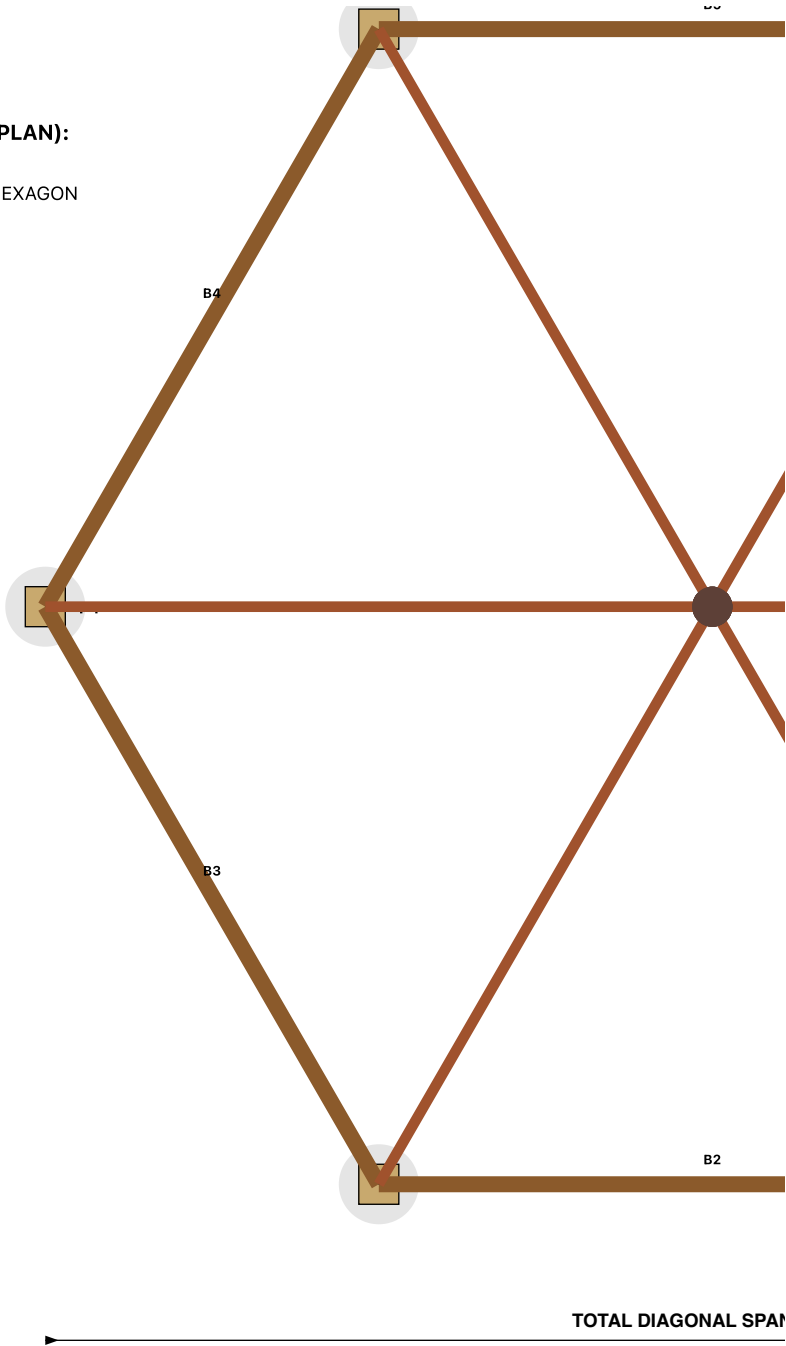


Perspective View

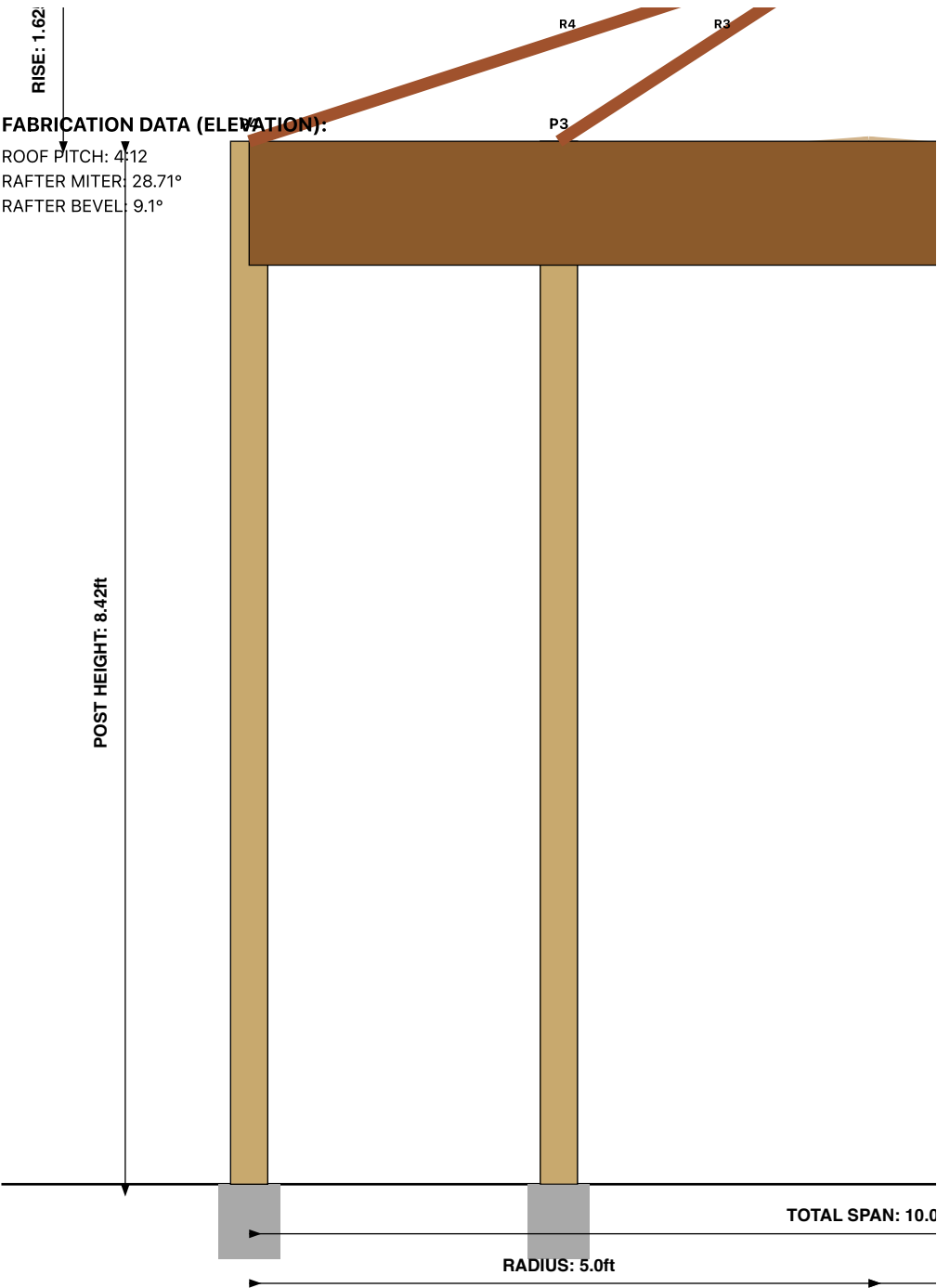
3D PERSPECTIVE MODEL
VISUALIZATION OF STRUCTURAL LOAD PATHS



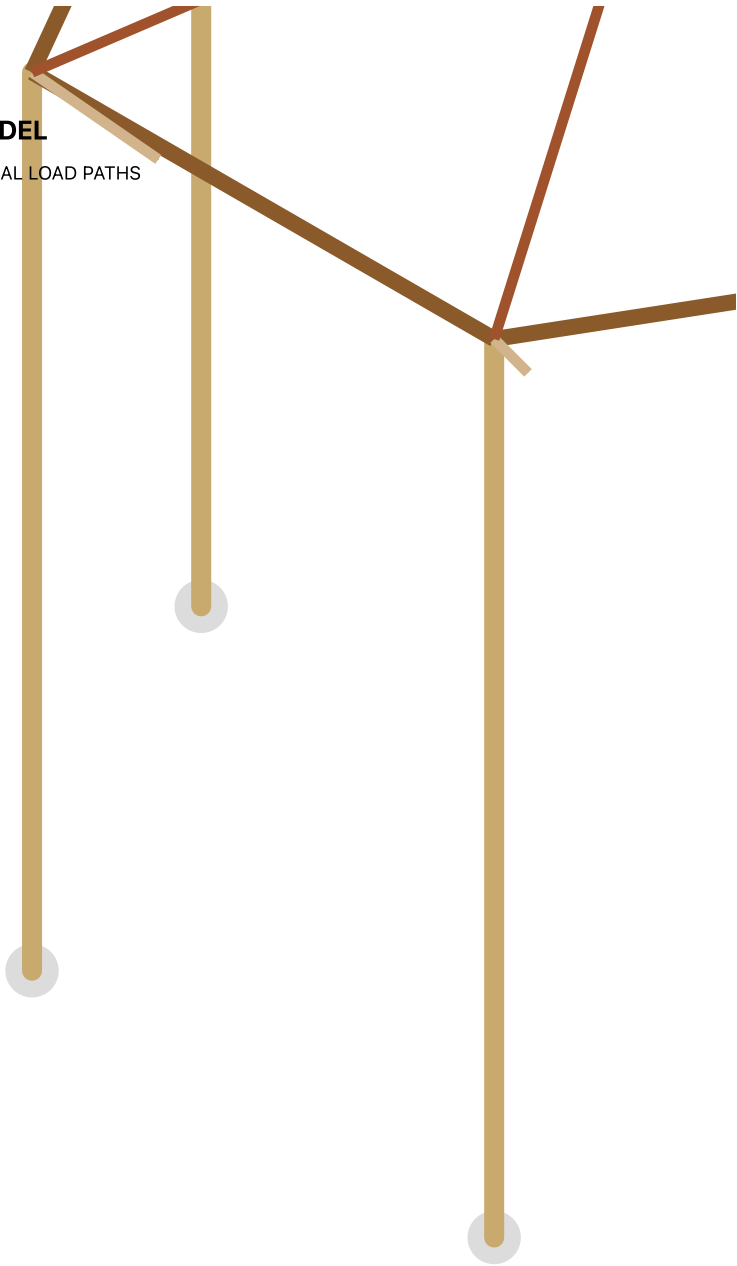
FABRICATION DATA (PLAN):
BEAM RING MITER: 30.0°
LAYOUT: EQUILATERAL HEXAGON



SB-02 Elevation

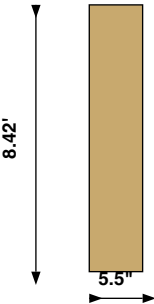


3D PERSPECTIVE MODEL
VISUALIZATION OF STRUCTURAL LOAD PATHS



A: POST + BASE DETAIL

PROFILE: 6x6 Cedar
CUT LENGTH: 8.42ft



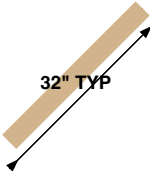
B: BEAM RING DETAIL

PROFILE: 2x8 S4S
RING MITER: 30.0°



D: KNEE BRACE CONNECTION

DETAIL: 4x4 KNEE BRACE



E: FOOTING ANCHORAG

DETAIL: 12" PIER

